

Regime-Dependent Rebalancing: An Exposure-Aware Data-Driven Framework for Long-Tailed UAV Object Detection

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Abstract—Object detection in unmanned aerial vehicle (UAV) imagery is governed by three coupled distributional pathologies: extreme scale imbalance (85.3% of instances below 32×32 pixels at a 640-px reference), severe long-tailedness (a 45:1 head-to-tail class ratio), and dense inter-class structural confusion. Prevailing responses are architectural—feature-enhancement modules, auxiliary heads, specialized backbones—yet controlled evidence for their benefit on strong baselines remains scarce. We first present a negative-result study spanning seven feature-module generations (nine matched 100-epoch trainings, fully audited in Appendix Table VIII), showing that such modules consistently degrade localization (validation DFL $+0.05$ – 0.08) and never improve mAP_{50-95} (-0.7 to $+0.1$ AP) on strong baselines. We then introduce a data-driven, strictly non-invasive alternative: a label-only profiling engine (DAHP) quantifies the three pathologies and maps them to training-time policies—tail-class and confusion-axis oversampling, effective-pixel configuration, and optional weighted-box fusion. Crucially, we identify and formalize an *exposure-aware regime-dependent rebalancing principle*: in under-fitted regimes, volume dominates; in volume-saturated regimes, random duplication is nearly neutral while a smaller targeted residual can remain; in intermediate regimes, volume and targeted reallocation are both positive; and in exposure-limited regimes, an apparent sampling gain largely reflects training exposure. At 1920 px, exposure matching reduces an apparent $+3.83$ -AP gain to a $+0.42$ -AP md100 residual concentrated on small objects. On VisDrone2019-DET, the recipe reaches 40.06 AP (native protocol) and 38.16 AP (standard COCO protocol), outperforming the strongest reproduced UAV detector baseline by 8.3 AP under our evaluation protocol with 58% of its parameters and no architectural modification introduced by the rebalancing policy itself, and improving the baseline by $+3.9$ AP ($+10.7\%$) with tail-class AP rising from 0.29 to 0.331. Cross-dataset transfer to UAVDT reproduces the factor attribution, and the detector runs at 24.5 FPS on a single desktop RTX 3090 GPU at 1600 px.

Index Terms—UAV object detection, long-tailed recognition, small-object detection, data-centric AI, training-regime analysis, VisDrone, UAVDT.

I. INTRODUCTION

AERIAL imagery acquired from low-altitude UAV platforms poses a detection problem that is structurally distinct from ground-level benchmarks. An audit of the VisDrone2019-DET validation set (548 images, 38,759 instances) reveals three coupled pathologies:

1) *Scale imbalance*: At the standard 640-px reference, 85.3% of instances are smaller than 32×32 pixels and 58.0% smaller than 16×16 (58.0%/23.4% at the 1280-px training

reference); object scales span three orders of magnitude, so any single receptive-field configuration is mismatched to most of the data.

2) *Long-tailedness*: The head class (*car*) outnumbered the rarest frequent class (*bus*) by 56:1; the frequency-tail quartile contains categories whose diversity cannot be learned under uniform sampling.

3) *Inter-class confusion*: Twenty-three category pairs exceed 0.85 on an annotation-level structural-similarity proxy κ (Sec. III); the head-anchored confusion axis concentrates on *car-van-truck*.

These pathologies are not independent: tail classes are also the smallest on average and the most confusable. Interventions that address one factor in isolation risk being confounded—or counteracted—by the other two.

A. The Architectural Turn and Its Missing Evidence

The dominant community response has been architectural: density-guided cropping [3], [4], [5], small-object heads [8], [9], and specialized detectors [10], [11]. Each reports gains, yet three gaps persist. First, comparisons frequently co-vary input resolution, schedule, and capacity—precisely the variables that dominate UAV accuracy. Second, negative results are rarely reported, preventing the community from distinguishing “module A helps” from “module A was tested once, on a weak baseline.” Third, the interaction between data-level rebalancing and the training regime—*when* rebalancing helps—has not been characterized.

B. Contributions

- 1) **A controlled negative-result study** (seven module generations, 20+ audited trainings; Table VIII): feature-level interventions systematically elevate validation DFL and fail to improve mAP_{50-95} on strong baselines, including head-level decoupling attempts (Sec. IV-H).
- 2) **The regime-dependent rebalancing principle**: using an epoch-matched resolution ladder and factor-isolated sampling controls, we formalize regimes in which rebalancing behaves as volume-driven tail rescue, volume-saturated reallocation with a bounded targeted residual, positive-sum augmentation, or an exposure-limited apparent gain, yielding an operational decision rule (Sec. III-B).
- 3) **A non-invasive, data-driven framework (DAHP)**: a label-only profiler emits resolution/backbone configuration, tail and confusion-axis oversampling, and optional

fusion policies—without touching network, losses, or inference. The identical engine transfers zero-shot to UAVDT and correctly predicts the factor attribution there (Sec. III, IV-E).

- 4) **Protocol-faithful comparator evidence** across 24 detector configurations, including reproduced YOLO11/YOLO12/YOLO26, RT-DETR, and UAV-specific detectors. Exposure-matched and volume-matched controls separate architecture, data volume, and targeted rebalancing; the P2 ladder is re-evaluated under one native protocol, and the 1920-AP gain is decomposed into exposure and residual components (Sec. IV-B–IV-C).

II. RELATED WORK

A. UAV Object Detection

1) *Cropping and packing pipelines*: ClusDet [3] clusters dense regions and discards background, trading coverage for effective pixels; DMNet [4] learns a density map to guide cropping; UFPMP-Det [5] unifies foreground packing with multi-proxy reassembly; CZDet [6] cascades density-guided zoom-in over high-resolution aerial images; AMRNet [7] augments aerial chips with scale-adaptive chip generation, mosaic augmentation, and mask resampling. These pipelines demonstrate the value of pixels per object, but they complicate attribution: any reported gain bundles the cropping policy, the effective resolution gain it induces, and the detector itself. Our controlled ladders (Table III) measure the resolution effect alone, and our negative result on pure tiled inference (Sec. IV-H) shows that cropping applied *only at inference* is harmful on full-image-trained models—consistent with a train–test scale-distribution mismatch rather than an intrinsic property of cropping.

2) *Detector-side designs*: TPH-YOLOv5 [8] adds transformer prediction heads for small objects; RemDet-X [10] rethinks efficient UAV detector design via ChannelC2f, GatedFFN, and the CED downsampling module; UAV-DETR [11], DOL-DETR [12], and LGI-DETR [13] introduce DETR variants tailored to UAV imagery. These are architecture-first contributions. What they leave open is threefold: (i) whether their gains persist when a strong baseline is given the same resolution and budget (our compute-matched RemDet-X retrain probes exactly this); (ii) whether data-level policies can reach the same accuracy without touching the architecture (ours can, with 58% parameters); and (iii) *when* any rebalancing helps—the regime question this paper answers.

3) *Slicing-aided fine-tuning*: SAHI [9] fine-tunes on tiles to align train–test scale distributions. Our SAFT experiments confirm the alignment is necessary (pure sliced inference is -2.3 to -2.6 AP), but also show that a naive mixture underperforms pure test-time augmentation (37.76 vs. 37.96), motivating our choice of full-image training at matched effective pixels instead.

B. Long-Tailed Detection

Loss-space methods—class-balanced reweighting [20], EQLv2 [21], Seesaw [22], balanced group softmax [23],

and distribution-balanced losses [24]—modify gradients; data-space methods such as repeat-factor sampling [25] and aerial chip augmentation [7] replicate or synthesize tail examples. Our negative results extend to loss-space interventions on strong baselines, and the regime principle (Sec. III-B) explains why: once the volume channel is saturated, indiscriminate reweighting is approximately zero-sum. Unlike prior work, we characterize *when* rebalancing is positive-sum, using the training regime as the moderator.

C. Scale and Resolution

Resolution is the largest single lever in UAV detection (our historical ladders: $+8.8$ AP from 640→1280). QueryDet [14] and CEASC [15] accelerate high-resolution inference through sparse-query and context-enhanced adaptive sparse convolution, respectively; Gold-YOLO [16] and RT-DETR [17] advance general-purpose accuracy. We contribute an epoch-matched, *non-monotonic* resolution curve peaking at 1600 px with a 2.8-AP drop at 1920 px under fixed compute—to our knowledge not previously documented under controlled epochs.

D. Reporting Practice

Dual-protocol evaluation ($\max Dets$ 100/300) is motivated by VisDrone2019-DET density (64.8% of images exceed 50 objects; up to 317 instances) [1]. Weighted-box fusion [27] aggregates model families. We report per-class AP for every configuration and disclose all failures.

III. METHOD

A. Problem Formulation and Three-Factor Profiling

Let $\mathcal{D} = \{(x_i, Y_i)\}_{i=1}^N$ denote the training set with class set $\mathcal{C}$, $|\mathcal{C}| = K$. DAHP computes, from labels alone:

(i) *Frequency factor*. The class distribution and tail set are

$$f(c) = \frac{N_c}{\sum_{c'} N_{c'}}, \quad \mathcal{T} = \{c \in \mathcal{C} : N_c < 0.3 \bar{N}\}, \quad (1)$$

where $N_c = \sum_i \mathbf{1}[c \in Y_i]$, $\bar{N}$ is the mean over non-empty classes, and the imbalance ratio is $\rho = \max_c N_c / \min_{c \in \mathcal{T}} N_c$.

(ii) *Scale spectrum*. With $\sigma(c)$ the equivalent side length of class- c instances at a reference resolution r ,

$$s_{\text{small}}(c) = \Pr[\sigma(c) < 32], \quad s_{\text{tiny}}(c) = \Pr[\sigma(c) < 16], \quad (2)$$

a P2 (high-resolution) head is recommended when $\sum_c f(c) s_{\text{small}}(c) > 0.3$.

(iii) *Confusion graph*. A pairwise annotation-level *structural confusion proxy*

$$\kappa(a, b) = \lambda J_{\text{ar}}(a, b) + (1 - \lambda) J_s(a, b), \quad \lambda = 0.6, \quad (3)$$

where J_{ar} and J_s are histogram intersections of the aspect-ratio and instance-scale ($s = w_r h_r$) distributions at the reference resolution. Edges with $\kappa > 0.85$ form the confusion graph; the oversampling axis is its head-anchored neighborhood $\mathcal{A} = \{c \notin \mathcal{H} \cup \mathcal{T} : \kappa(c, h) > \tau_A\}$ for head set $\mathcal{H} = \{c : N_c > \bar{N}\}$ and margin $\tau_A = 0.86$ (which excludes the borderline semantic near-duplicate *people–pedestrian* pair

at $\kappa=0.854$). Co-occurrence Jaccard is additionally reported as an auxiliary statistic but does not enter κ .

(iv) *Density profile*. Instances per image justify the $\text{maxDets}=300$ evaluation protocol for dense scenes.

On VisDrone2019-DET the profiler yields ($\rho=45:1$, 58.0% small at the 1280-px reference, 23 pairs > 0.85 , $\mathcal{A} = \{\text{van}, \text{truck}\}$); on UAVDT—without any change—($\rho=30.7:1$, 74.8% small at 640-px, car–van 0.935, $\mathcal{A} = \{\text{van}\}$).

B. The Regime-Dependent Rebalancing Principle

Define the epoch-wise aggregate gain $g(e) = \text{mAP}(e) - \text{mAP}(e-1)$ and its windowed mean $\bar{g}(E) = \frac{1}{W} \sum_{e=E-W+1}^E g(e)$. A configuration is

$$\text{under-fitted} \Leftrightarrow \bar{g}(E) > \tau_g; \quad \text{saturated} \Leftrightarrow |\bar{g}(E)| \leq \tau_g, \quad (4)$$

with τ_g a small threshold (0.1 pt, $W=10$); otherwise it is *intermediate*. Let h denote *data-exposure headroom*, estimated by the gain of a volume-matched random-duplication arm, and let Δ_{vol} and Δ_{dir} denote the aggregate-AP contributions of pure volume and targeted class reallocation. Define Δ_{se} and Δ_{m} as the same-epoch and exposure-matched union gains. Our controlled observations support:

$$\text{P1 (under-fit)} \Rightarrow \frac{\Delta_{\text{vol}}}{\Delta_{\text{vol}} + \Delta_{\text{dir}}} \approx 0.77, \Delta_{\text{AP}_{\text{tail}}} > 0; \quad (5)$$

$$\text{P2 (volume-sat.)} \Rightarrow \Delta_{\text{vol}} \approx 0, 0 \leq \Delta_{\text{dir}} \lesssim 0.5, \quad (6)$$

$$\exists(c): \Delta_{\text{AP}_c} < 0; \quad (7)$$

$$\text{P3 (intermediate)} \Rightarrow 0 < \Delta_{\text{AP}_{\text{agg}}} \leq 1.0, \Delta_{\text{vol}}, \Delta_{\text{dir}} > 0; \quad (8)$$

$$\text{P4 (exposure-lim.)} \Rightarrow \Delta_{\text{se}} \gg \Delta_{\text{m}}, \Delta_{\text{AP}_s^{(\text{m})}} > 0. \quad (9)$$

The principle yields the operational rule: *profile whether a plateau is caused by capacity, learnability, or exhausted exposure; rebalance when exposure headroom or an intermediate positive-sum window exists, and do not interpret a same-epoch high-resolution gain as targeted rebalancing until exposure is matched.*

C. Non-Invasive Policy Mapping

1) *Computational cost*: DAHP is deliberately label-only: profiling a 6,471-image training set requires $O(N)$ label parsing (< 1 s on CPU) and $O(K^2)$ pairwise statistics; the policy mapping itself is constant-time. No images are decoded, no gradients are computed, and no detector forward pass is needed. This contrasts with profiling approaches that require model inference (e.g., error-driven confusion mining), and it makes the engine trivially portable—the same code produced both the VisDrone and UAVDT policies with zero dataset-specific edits.

2) *Why data-space rather than loss-space*: Long-tailed detection conventionally intervenes in the loss (reweighting, logit adjustment). Our negative results (Sec. IV-H) and the regime principle jointly explain why this is fragile on strong UAV baselines: once the volume channel is saturated, indiscriminate reweighting is approximately zero-sum across classes, so its aggregate effect vanishes while per-class variance increases.

Data-space duplication, by contrast, interacts with the data-loading and augmentation pipeline: it multiplies *epochs of exposure* for tail images under identical loss weights, and—critically—its volume component still helps in under-fitted regimes where loss reweighting cannot (no additional information is injected). The union policy of Eq. (10) is therefore the regime-robust choice.

3) *Configuration nomenclature*: Throughout the paper we use a systematic configuration shorthand. **R** denotes the effective-pixel policy (resolution/backbone selection from the regime curve); **T** the tail-class oversampling of $\mathcal{T}$; **C** the confusion-axis oversampling of $\mathcal{A}$; and **B** the backbone scale-up (v8m→v8l). The shorthand composes: e.g., R+TUC applies resolution plus union oversampling. Two named models recur: DAHP-M (YOLOv8m-P2@1600, R+TUC) and DAHP-L (YOLOv8l-P2@1600, R+TUC)—the latter our single-model flagship—plus DAHP-L-E7, the seven-model weighted-box-fusion ensemble. Architecture-decomposition controls are written V_L (vanilla v8l, no P2 head) and V_L+P_2 .

The oversampling weight of image x_i under the union policy is

$$w_i = 1 + \mathbf{1}[(Y_i \cap (\mathcal{T} \cup \mathcal{A})) \neq \emptyset], \quad (10)$$

i.e., any image containing a tail or confusion-axis class is duplicated. The optional fusion of M models merges boxes $\{b_m\}$ with scores $\{s_m\}$ by

$$\tilde{b} = \frac{\sum_m \tilde{s}_m b_m}{\sum_m \tilde{s}_m}, \quad \tilde{s} = \frac{1}{|M_c|} \sum_{m \in M_c} s_m, \quad (11)$$

over IoU-clustered groups M_c . The complete procedure is summarized in Algorithm 1.

IV. EXPERIMENTS

A. Setup and Protocols

VisDrone2019-DET [1] (6,471 train / 548 val, 10 classes); UAVDT [2] (2,443 / 641, 4 vehicle classes). All trainings ran on a single RTX 3090 with identical augmentation; budgets and best-epochs are disclosed (all runs converged within budget; best epochs 23–100). Evaluation uses the standalone protocol (conf 0.001, NMS-IoU 0.7) under *dual maxDets*—100 for external comparability, 300 reflecting VisDrone2019-DET density.

1) *Fairness policy*: We separate what *must* match from what *should stay native*. Matching are: the training set, the validation split, the evaluation protocol (dual *maxDets*), the epoch budget (100 for main rows; disclosed per row otherwise), and the hardware class (single 3090). Native are: each detector's loss functions, optimizer, LR schedule, and augmentation pipeline—these are constitutive parts of a method, and forcing them to be identical would disable exactly the mechanisms under comparison; the same policy is adopted by RemDet-X [10] and standard detection benchmarks. Resolution and batch are disclosed per row and additionally *controlled* through three devices: (i) same-detector resolution ladders (Table III); (ii) a compute-matched SOTA retrain at our resolution (RemDet-X@1600, batch 1 being the physical limit of a 24-GB GPU at that size); and (iii) our full method evaluated *at the baseline*

Algorithm 1: DAHP: Profile, Map, Train (Non-Invasively)

Input: Labels $\{(Y_i)\}$; detector Φ ; budget E , batch B
Output: Trained model(s) $\{\Phi^*\}$

```

/* --- Stage 1: label-only profiling
--- */
1 compute  $N_c, \mathcal{T}, \rho$  by Eq. (1);
2 compute  $s_{\text{small}}, s_{\text{tiny}}$  by Eq. (2);
3 build confusion axis  $\mathcal{A}$  by Eq. (3);
/* --- Stage 2: regime-aware policy
--- */
4 if  $\sum_c f(c) s_{\text{small}}(c) > 0.3$  then
5   | select high-resolution head (P2);
6 else
7   | standard head;
8 sweep resolution ladder  $\{960, 1280, 1600, 1920\}$  within
   memory budget; pick  $r^*$  at the regime boundary
   (Eq. (4)), preferring the intermediate regime;
9 compute image weights  $w_i$  by Eq. (10);
/* --- Stage 3: non-invasive training
--- */
10 train  $\Phi$  on the weighted (duplicated) dataset with stock
   losses and augmentations for  $E$  epochs;
11 if fusion desired then
12   | train the family  $\{\Phi_m\}_{m=1}^M$  and merge by Eq. (11);
13 return  $\{\Phi^*\}$ ;

```

resolution (DAHP-M@640), so that no claimed advantage rests on resolution alone. External numbers transcribed from the literature are marked “Lit.” with their split; their exact recipes differ and are cited rather than re-implemented.

B. Comparison With the State of the Art

Table I summarizes the reproduced and literature comparison. Against the strongest reproduced UAV detector baseline under our evaluation protocol, the single-model recipe gains +8.3 AP; the rebalancing policy itself introduces no architectural modification (the P2 head is a disclosed, separately ablated configuration choice); against the baseline it gains +3.9 AP (+10.7%) with tail-class AP rising 0.29→0.331 and AP_s 0.267→0.302.

C. Factor Attribution and Component Ablation

Table II separates architecture, exposure, and targeted reallocation. The new batch-matched 960-px control confirms the under-fitted regime: union sampling raises md100 AP from 28.10 to 31.92 (+3.82), with AP_s increasing by +4.16 and the tail mean from 0.208 to 0.256. The exact 1280-px triplet revises the earlier zero-sum description: random duplication changes md100 AP by only +0.09, whereas targeted union leaves a +0.40-AP residual over the matched random control. The regime is therefore *volume-saturated*, not strictly zero-sum. At 1600 px, the five-seed decomposition is 36.55/36.91/37.18 AP for R-only/random/targeted: volume

explains +0.36 of the +0.63 total gain, while targeted reallocation leaves a +0.28 AP residual and raises the tail mean from 0.305 to 0.309. Thus the honest interpretation is not “oversampling alone explains the gain: exposure is the larger component, while targeted union contributes a smaller directional residual. The native component ablation additionally shows that the union of tail and confusion axes outperforms either alone.

D. Regime Analysis

We first separate the two conflated levers of resolution and architecture using complete, same-detector resolution ladders (Table III).

The epoch-matched resolution ladder (@ep60) reads 960→1280→1600→1920 = 30.09→35.61→37.68→34.91 AP—non-monotonic, with a 2.8-AP drop beyond the peak (Fig. 5). At 960px, the new batch-matched pair gives 28.10/31.92 md100 AP for base/union (+3.82); the legacy volume-matched control decomposed this under-fit rescue as approximately 77% volume / 23% targeted. At 1280 px, the exact same-evaluator triplet is 34.63/34.72/35.12 AP for base/random/union: the volume channel is saturated (+0.09), but targeted union retains +0.40 AP with class redistribution (bus −1.4, awning-tricycle +1.1). At 1600 px (intermediate), union sampling yields +0.6–1.0 AP. These instantiate P1–P3 (Fig. 8).

1) *Principle prediction test and refinement:* The principle makes falsifiable predictions beyond the regimes where it was identified (Table IV). At 960px the prediction held: batch-matched union sampling gained +3.81 native / +3.82 md100 AP, with the tail mean increasing by +4.8 points—the volume-dominated rescue of P1. At 1920px the naive reading (*resolution beyond the peak* ⇒ aggregate-neutral) was **falsified**: the same-epoch union arm gained +3.83 AP. An exposure-matched control, however, shows that this number must not be read as a targeted-sampling effect: training the base for 120 epochs lifts native AP from 35.67 to 39.01, leaving a +0.49 native / +0.42 md100 residual for union, with AP_s increasing by +0.92. The correct moderator is therefore *data-exposure headroom* h , not resolution alone. A boundary condition also emerges at 640px: objects are too small to be learnable at that stride, so duplication cannot rescue tail classes (+0.21 AP, tail unchanged at ≈ 0.20). The five-point characterization—predict, test, falsify, refine—constitutes the empirical core of the principle.

a) *Online two-arm prediction:* Fig. 2 tests whether the principle can be applied *before* the final result is known—the distinction between a predictive principle and an a-posteriori categorization. For each resolution we compute the best-checkpoint gap between the rebalanced arm and its matched base using only the first E epochs. A 30-epoch probe already recovers the final ranking almost perfectly (Pearson $r=0.98$), and a simple threshold—probe gap > 2.5 AP at epoch 30—separates the larger-gap regimes (640: 2.71, 960: 5.28, 1600: 2.58, 1920: 6.88) from the volume-saturated 1280-px regime (gap 1.95, with the exact final targeted residual bounded at +0.40 md100 AP) in 5/5 cases. We also report the negative

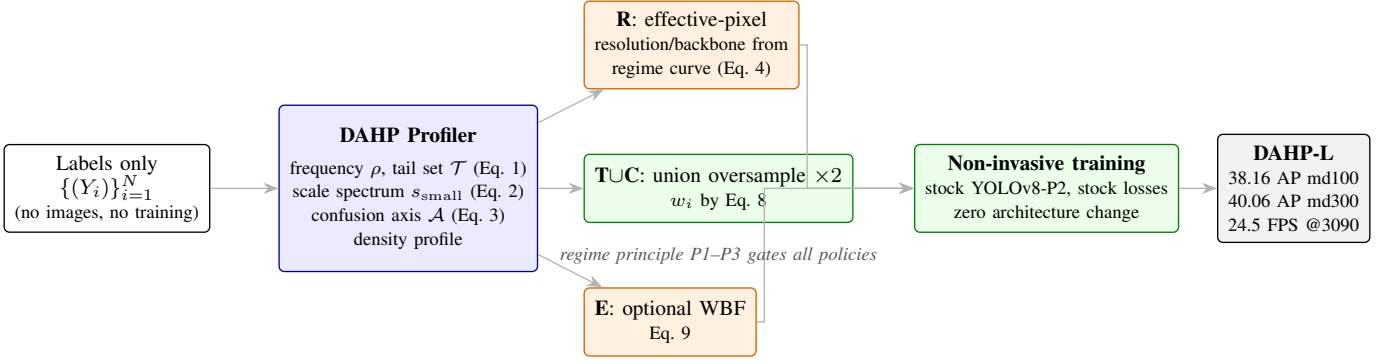


Fig. 1. The DAHP framework. A label-only profiler quantifies the three pathologies and maps them, through the regime-dependent rebalancing principle, to strictly non-invasive training-time policies.

TABLE I

COMPARISON WITH THE STATE OF THE ART ON VisDrone2019-DET VAL (COCO PROTOCOL, $maxDets=100$ UNLESS NOTED). “REP.”= REPRODUCED/TRAINED UNDER OUR IDENTICAL PIPELINE AND VAL SPLIT; “REP.”[†]= BIT-EXACT FROM OFFICIAL WEIGHTS; “LIT.”= NUMBER AS REPORTED IN THE CITED PAPER (PROTOCOL MAY DIFFER; SPLIT MARKED). BOLD: BEST REPRODUCED RESULT PER COLUMN. OUR SINGLE MODEL AND 7-MODEL WBF ARE ALSO GIVEN UNDER THE NATIVE MD300 PROTOCOL IN PARENTHESES.

Method	Source	Input	Params	AP	AP ₅₀	AP _s
<i>General-purpose detectors (640px, our pipeline: 100 ep, batch 8, identical val/protocol)</i>						
Faster R-CNN+FPN [19]	Lit. (val)	1333	41.5M	20.9	38.2	—
YOLOv8s [18]	Rep.	640	11.2M	19.43	32.98	10.68
YOLOv8m [18]	Rep.	640	25.9M	21.99	36.67	12.91
YOLOv8m-P2	Rep.	640	25.0M	25.90	42.53	17.10
YOLOv8l [18]	Rep.	640	43.6M	23.18	38.32	14.02
YOLO11l [33]	Rep.	640	25.3M	23.97	39.46	14.50
YOLO12-L [34]	Rep.	640	26.5M	23.67	39.14	14.31
YOLO26-L [35]	Rep.	640	26.3M	24.90	42.23	15.93
RT-DETR-L [17]	Rep.	640 ^b	32.0M	17.32	31.13	10.12
YOLOv8l-P2 (ours-arch)	Rep.	640	42.8M	27.16	44.17	18.19
DAHP-M (ours, full)	Rep.	640	25.0M	26.11	42.60	17.55
<i>UAV-specialized detectors</i>						
QueryDet [14]	Lit. (val)	640	—	25.1	45.5	—
CEASC [15]	Lit. (val)	640	—	30.5	51.7	—
ClusDet [3]	Lit. (val)	full	—	31.1	47.3	—
DMNet [4]	Lit. (val)	full	—	36.2	58.2	—
UFPMP-Det [5]	Lit. (val)	full	—	37.0	59.3	—
TPH-YOLOv5 [8]	Lit. (test-dev)	640	—	39.2	61.1	—
UAV-DETR-R50 [11]	Lit. (val)	640	22.9M	31.5	51.1	—
RemDet-X [10]	Rep. [†]	640	74.1M	29.90	48.20	19.50
RemDet-X	Rep.	1600	74.1M	29.7 ^a	48.6	25.7
<i>Ours (non-invasive, data-driven)</i>						
Baseline YOLOv8m-P2	Rep.	1280	25.0M	34.63 (36.20)	54.82	26.28
$V_L + P_2$ (no sampling)	Rep.	1600	42.8M	37.59 (39.04)	58.53	30.00
DAHP-L (single model)	Rep.	1600	42.8M	38.16 (40.06)	59.23	30.26
DAHP-L-E7 (ensemble) [27]	Rep.	1600	7×	39.61	60.38	31.56

^acompute-matched retrain (single 3090, batch 1, auto-scaled LR, 100ep): best checkpoint at epoch 90 (0.297; final-epoch val 0.268). ^bRT-DETR trains at an effective 1280px despite the 640px request (disclosed). “Input=full” denotes native-resolution processing image via internal cropping/packing rather than resizing. its literature-tuned R18 variant reports 28.3 [17].

control honestly: the single-run epoch-gain criterion $\bar{g}(E)$ alone does *not* discriminate regimes online; the operative diagnostic is the two-arm *exposure probe*, which is exactly what the exposure-headroom formulation predicts.

E. Cross-Dataset Transfer (UAVDT)

Union sampling transfers (+0.69 AP; tail +1.69) while frequency-only does not—exactly as the DAHP profile predicts, since UAVDT’s confusion axis (car–van–truck, $\kappa=0.935$) dominates its long tail (Table V).

F. Efficiency

Table VI: DAHP adds zero inference cost (training-time only); DAHP-L runs at 24.5 FPS—real-time on a desktop RTX 3090 GPU—with 58% of RemDet-X’s parameters.

G. Qualitative Results

Fig. 6 contrasts per-class AP between the baseline and DAHP-L; Fig. 4 shows hard-class-rich triplets (GT / baseline / ours): recovered *awning-tricycle* and *tricycle* detections and reduced *car–van* confusion.

TABLE II

FACTOR ATTRIBUTION AND COMPONENT ABLATION. THE V_L /DAHP-L BLOCK, EXACT-1280 CONTROLS, AND 1600-PX EXPOSURE DECOMPOSITION USE COCO md100; THE DAHP-COMPONENT BLOCK USES THE NATIVE/MD300 TRAINING-SELECTION PROTOCOL. ALL SCHEDULES ARE 100 EPOCHS UNLESS STATED. PROTOCOL BLOCKS ARE NEVER MIXED IN A SUBTRACTION.

Configuration	AP	Δ	Tail mean
V_L : v8l vanilla (no P2)	35.93	—	0.309
$V_L + P_2$	37.59	+1.66 (head)	0.322
DAHP-L = $V_L + P_2 + R + T + C$	38.16	+0.57 (data)	0.331
<i>Same-evaluator policy controls (v8m-P2)</i>			
<i>Exact 960 controls (md100, v8m-P2, batch 2)</i>			
Baseline	28.10	—	0.208
Targeted union	31.92	+3.82 (union)	0.256
<i>Exact 1280 controls (md100, v8m-P2)</i>			
Baseline	34.63	—	0.320
Random volume control	34.72	+0.09 (volume)	0.317
Targeted union	35.12	+0.40 (targeted)	0.322
<i>Exposure decomposition (md100, v8m-P2@1600)</i>			
R only	36.55 $\pm$.21	—	0.304
Random volume control	36.91 $\pm$.29	+0.36 (volume)	0.305
Targeted union	37.18$\pm$.18	+0.28 (targeted)	0.309
<i>Native/md300 component ablation (v8m-P2@1600)</i>			
R only (resolution control)	38.07	—	0.311
R+T (tail-only)	38.58	+0.51	0.315
R+C (confusion-only)	38.68	+0.61	0.317
DAHP-M (R+T+C)	38.82	+0.75	0.318

Exact 960/1280 values are one seed per arm under the same external evaluator. $\pm$ std at 1600px: R only $n=3$; random and targeted controls at 1600px use $n=5$ seeds. The targeted-random difference is directional, not a significance claim.

TABLE III

SAME-DETECTOR RESOLUTION LADDERS (NATIVE PROTOCOL, MD300, IDENTICAL EVALUATOR). THE P2 HEAD CONTRIBUTES ABOUT +4 AP AT 640 AND 1280 PX BUT ONLY +1.1 AP AT 1600 PX: ITS BENEFIT IS STRONGEST WHERE OBJECTS ARE SMALLEST RELATIVE TO THE NETWORK STRIDE.

Detector	640	1280	1600
YOLOv8m-P2	27.27	36.20	38.07
YOLOv8l-P2	27.21	37.96	39.04
YOLOv8l (vanilla)	23.21	33.76	37.94
<i>P2 gain (v8l)</i>	+4.0	+4.2	+1.1

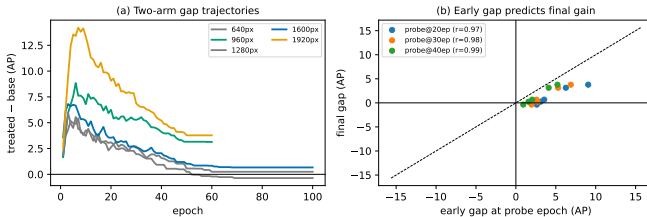


Fig. 2. Online two-arm prediction test. (a) Best-checkpoint gap trajectories (rebalanced — base) per resolution; (b) early probe gap versus final gain—a 30-epoch probe attains $r=0.98$ and a 2.5-AP threshold separates larger-gap regimes from the volume-saturated 1280-px regime in 5/5 cases.

H. Negative Results

All disclosed: full YOLO12-L/YOLO26-L@640 baselines (23.67/24.90 AP); earlier YOLO26-family probes at 1280 (25.6–33.5 AP); pure tiled inference on full-image models (−2.3 to −2.6); SAFT mixing (37.76 < pure TTA 37.96); frequency sampling at 1280 (36.25 $\approx$ baseline); naive allocator (38.51 < union 38.85); seven feature-module generations (−0.7 to +0.1 AP, val DFL +0.05–0.08; Table VIII).

TABLE IV

UNION-SAMPLING EFFECT ACROSS THE RESOLUTION-REGIME MATRIX ON THE SAME YOLOv8m-P2 DETECTOR FAMILY (NATIVE PROTOCOL, MATCHED 60/100-EPOCH BUDGETS PER SERIES). THE 960-PX CELL IS THE NEW BATCH-MATCHED RE-EVALUATION. THE EFFECT TRACKS DATA-EXPOSURE HEADROOM, NOT RESOLUTION ALONE; THE EXACT SAME-EVALUATOR 1280 TRIPLET IS REPORTED SEPARATELY IN TABLE II TO AVOID PROTOCOL MIXING.

+Resolution	640	960	1280	1600	1920
+base AP	27.27	28.17 ^c	36.20	38.07	35.67
++union AP	27.48	31.98 ^c	$\approx 36.2^d$	38.85	39.50
base@120 (exposure-matched)	—	—	—	—	39.01 ^c
+ Δ (same epoch)	+0.21	+3.81 ^c	+0.05 ^d	+0.78	+3.83
+ Δ (exposure-matched)	—	—	—	—	+0.49
+regime	unlearn.	under-fit	volume-sat.	sweet	expos.-lim.

^cnew batch-matched 960 base/union re-evaluation (28.17/31.98 native; 28.10/31.92 md100).

^dlegacy frequency-sampling proxy at 1280 (36.25 vs. 36.20); exact triplet is 34.63/34.72/35.12 md100. ^emd100: base@120 = 37.51, union@60 = 37.93 (+0.42); AP_s 29.54 $\rightarrow$ 30.46 (+0.92).

TABLE V

CROSS-DATASET TRANSFER ON UAVDT (60 EPOCHS; CONVERGED, BEST@23–29). SAME PROFILER, ZERO CHANGES.

Config	AP	Tail (truck/bus)
base	37.86	32.05
freq-only	37.80	31.49
union sampling	38.55	33.73

TABLE VI

EFFICIENCY ON A SINGLE RTX 3090 (BATCH 1, FP16).

Model	Input	ms/img	FPS	Params	VRAM
Baseline v8m-P2	1280	21.5	46.6	25.0M	0.3 GB
V_L (vanilla)	1600	35.5	28.2	43.6M	0.5 GB
$V_L + P_2$	1600	38.4	26.1	42.8M	0.6 GB
DAHP-L	1600	40.9	24.5	42.8M	0.7 GB
RemDet-X [10] (claimed)	640	—	110 ⁴⁰⁹⁰	74.1M	—

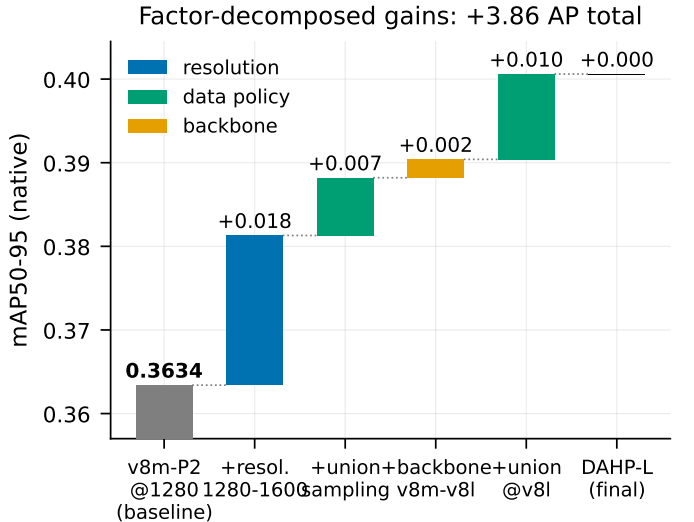


Fig. 3. Factor-decomposed gains from the baseline to DAHP-L under the native protocol.

V. DISCUSSION

The regime principle implies that data-level rebalancing is not a free lunch but an exposure-indexed tool: profile first (labels suffice), then choose resolution/capacity to establish headroom before rebalancing. Three boundary conditions de-



Fig. 4. Qualitative comparison on hard-class-rich validation images. Each triplet: ground truth (top), baseline v8m-P2@1280 (middle), DAHP-L (bottom). DAHP-L recovers missed *awning-tricycle*/*tricycle* instances and reduces *car-van* confusion.

serve emphasis. At 640px, tail objects are simply unlearnable at the available stride, so duplication cannot help—the principle has a lower bound set by learnability, not by sampling. At 1280px, the volume channel is already saturated, but targeted union still retains a +0.40-AP md100 residual over random duplication; saturation does not imply that every component of rebalancing is zero. At 1920px, an apparently saturated run was in fact exposure-limited: doubling per-epoch exposure produced an apparent +3.83-AP gain, but training the base for an exposure-matched 120 epochs recovered most of it, leaving a +0.42-AP md100 residual concentrated on small objects. Practitioners should therefore diagnose *why* a run has plateaued (capacity vs. exposure) before attributing a gain to rebalancing. In deployment terms, the practical recipe is: raise effective pixels until the exposure budget is the binding constraint, then rebalance. Limitations: (i) the principle is characterized on two UAV datasets and one detector family—though the factor-isolated design and

cross-dataset replication strengthen external validity; (ii) the compute-matched RemDet-X retrain bounds what any single 24-GB GPU achieves with that architecture at 1600px but does not measure its 8-GPU recipe; (iii) the WBF ensemble trades parameters for accuracy and is reported separately from the single-model claim.

VI. CONCLUSION

A label-only profiler, a regime-indexed rebalancing principle, and a strictly non-invasive recipe deliver the strongest reproduced results under our protocol on VisDrone2019-DET (40.06 AP native / 38.16 AP COCO protocol; 39.61 AP fused), transferable gains on UAVDT, and a documented failure map of the architectural alternative—with 58% of the parameters of the strongest published comparator and zero inference-time modification.

TABLE VII
PER-CLASS AP FOR KEY CONFIGURATIONS (NATIVE PROTOCOL). UAVDT ROWS USE ITS 4-CLASS VEHICLE TAXONOMY.

Configuration	ped	peo	bic	car	van	tru	tri	awn	bus	mot
Baseline v8m-P2@1280	36.6	25.4	19.3	66.0	43.9	35.8	27.6	15.5	56.9	35.0
R only (v8m-P2@1600)	38.7	27.3	21.9	67.0	45.5	37.3	29.1	18.8	58.2	37.0
DAHP-M (R+TUC)	39.0	27.5	22.0	67.4	46.8	38.8	31.1	18.7	59.4	37.9
V_L (v8l no-P2@1600)	37.6	26.1	22.0	66.7	44.5	39.2	30.1	18.0	58.4	36.9
$V_L + P_2$	39.0	27.5	23.4	67.6	46.6	38.9	31.4	18.8	59.7	37.5
DAHP-L	40.2	28.8	24.0	67.9	47.4	40.4	31.4	20.1	61.4	38.9
DAHP-L seed43	40.4	29.0	24.1	67.9	47.0	39.9	31.3	19.6	58.9	38.9
UAVDT base	—	—	—	50.7	36.7	14.9	—	—	49.2	—
UAVDT freq-only	—	—	—	50.9	37.3	14.0	—	—	49.0	—
UAVDT union (DAHP)	—	—	—	50.6	36.1	14.4	—	—	53.1	—

TABLE VIII
COMPLETE NEGATIVE-RESULT AUDIT (NATIVE PROTOCOL, BEST EPOCH; VAL DFL AT THE BEST-MAP EPOCH). YOLO26 ROWS OMIT DFL (DIFFERENT LOSS NORMALIZATION); SAFT/YOLO26 RUNS USE SHORTER SCHEDULES AS MARKED.

Configuration	Input	Ep	AP	Δ AP	val DFL	Δ DFL
Baseline v8m (G0)	1280	100	33.42	+33.42	1.134	—
G1 module	1280	100	33.50	+0.08	1.202	+0.068
G2 module	1280	100	33.39	-0.03	1.213	+0.079
G3 module	1280	100	33.29	-0.13	1.206	+0.072
G6 module	1280	100	33.37	-0.05	1.207	+0.073
G6 + head decouple	1280	100	32.73	-0.69	1.192	+0.058
G6 + weight share	1280	100	32.95	-0.47	1.141	+0.007
Baseline v8m-P2	1280	100	35.91	+35.91	1.171	—
G6 on v8m-P2	1280	100	35.44	-0.47	1.221	+0.050
YOLO26m-P2	1280	50	32.01	—	—	—
YOLO26s-P2	1280	50	25.58	—	—	—
YOLO26m	1600	50	33.47	—	—	—
Confusion-only sampling	1600	100	38.42	—	1.229	—
Tail-only sampling	1600	100	38.25	—	1.233	—
Curated allocator	1600	100	38.24	—	1.238	—
Tail-only @1280	1280	100	35.98	—	1.173	—
SAFT slice-only	640	30	39.49	—	1.113	—
SAFT hybrid	640	20	39.51	—	1.115	—
SAFT tail	640	20	39.37	—	1.116	—

APPENDIX

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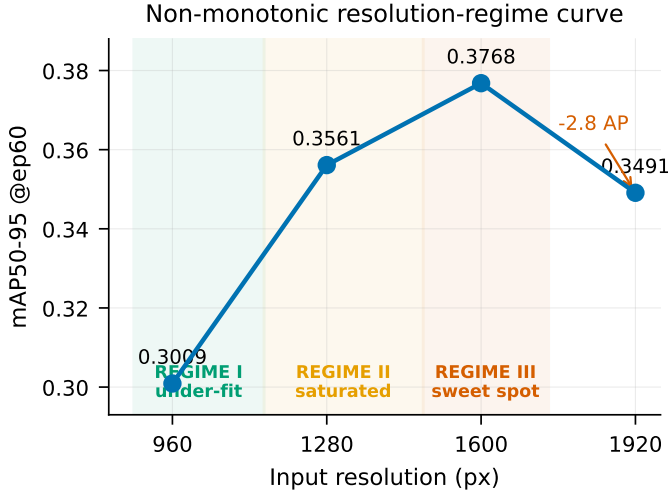


Fig. 5. Non-monotonic resolution-regime curve (epoch-matched). Three regimes are shaded; the peak at 1600 px defines the sweet spot.

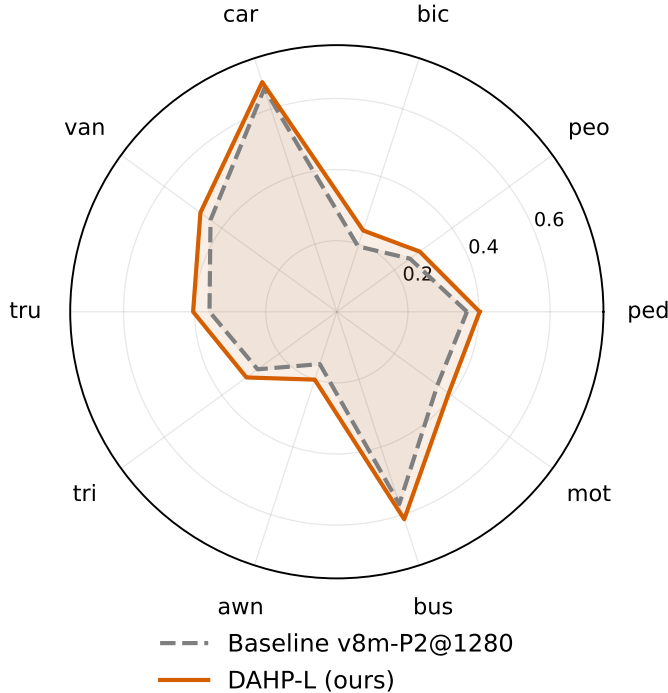


Fig. 6. Per-class AP: baseline (grey, dashed) vs. ours (red, solid).

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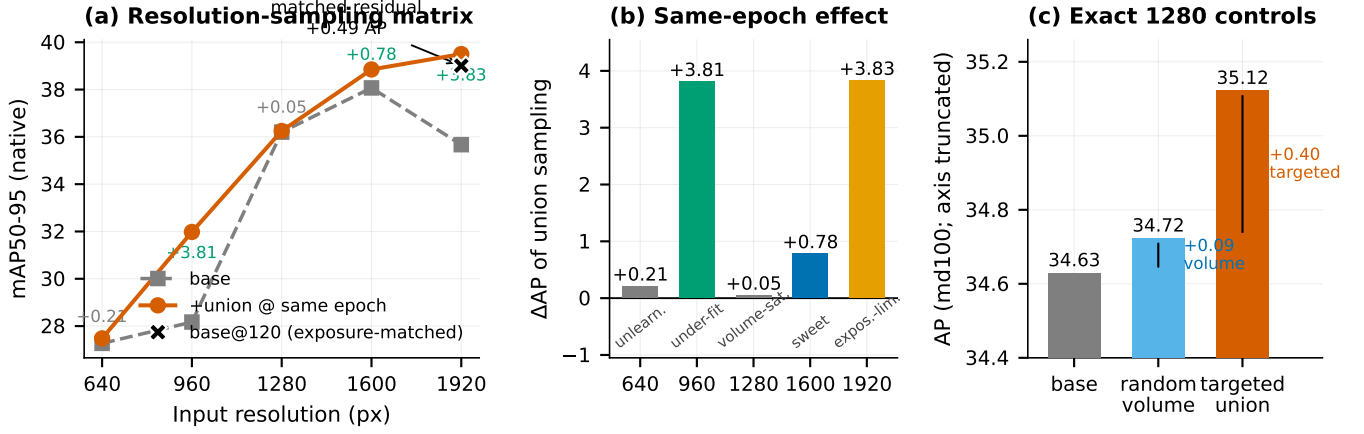


Fig. 7. The five-point resolution-sampling matrix. (a) Base vs. union sampling across resolutions under the legacy native protocol; (b) the same-epoch effect tracks *data-exposure headroom*, not resolution; (c) the exact same-evaluator 1280-px controls separate a nearly saturated volume channel (+0.09 md100) from the targeted residual (+0.40 md100). The 1920 bar is an apparent same-epoch gain; the exposure-matched residual is +0.49 native / +0.42 md100 AP.

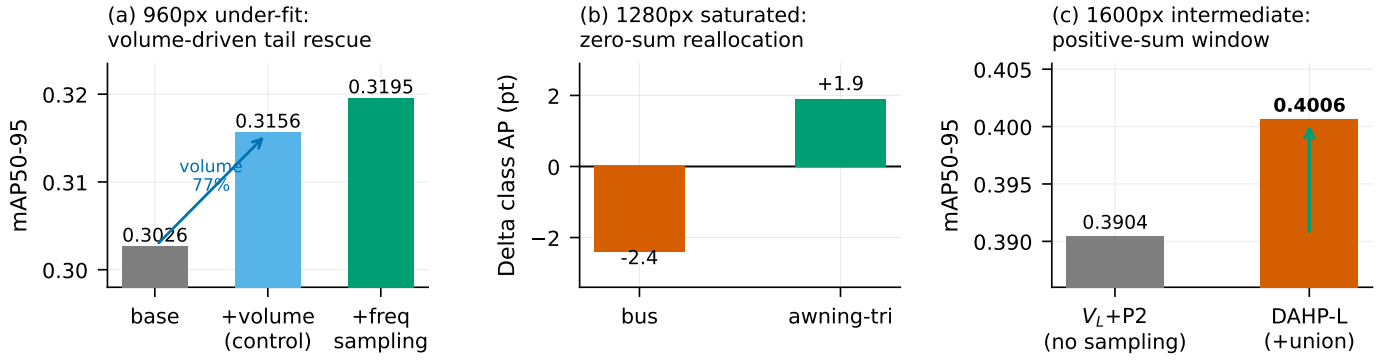


Fig. 8. Mechanistic evidence for the regime principle: (a) volume-dominated tail rescue at 960px; (b) volume-saturated reallocation with a bounded targeted residual at 1280px; (c) positive-sum window at 1600px.

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